

Multivariate bias correction improves inter-station dependence in multi-site daily precipitation without altering single-site performance: an evaluation of MBCn against quantile delta mapping over a coastal gauge network in Thailand

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Abstract

Statistical bias correction is applied to climate-model rainfall at almost every impact study, yet its performance is conventionally judged station by station. That practice cannot reveal what a correction does to the statistical relationships between gauges, which is precisely the property that distributed hydrological models and multi-site risk analyses depend on. This study asks how much a dependence-aware correction adds once single-site performance is held fixed. N-dimensional multivariate bias correction (MBCn) was evaluated against univariate quantile delta mapping (QDM) and uncorrected model output across twelve complete daily rain-gauge records on the western Gulf of Thailand coast, driven by seven CMIP6 models, trained on 1981–2000 and scored on a withheld 2001–2014 window. Because MBCn reinstates the QDM value at each gauge as its closing step, the two corrections cannot differ on any single-site statistic, and the six marginal scores confirmed this to three decimal places across all eighty-four gauge–model combinations. The methods separated on all six dependence diagnostics. MBCn lowered the multivariate energy distance by a third relative to QDM and cut the Spearman correlation-structure error from 0.286 to 0.187, down from 0.484 in uncorrected output. Each of the seven models improved on each diagnostic, giving the maximum attainable rank-biserial effect size and a one-sided Wilcoxon p of 0.0078. The iterative step converged within roughly fifteen passes, proved insensitive to the random seed, and preserved projected quantile changes as faithfully as QDM. Multivariate correction is therefore justified wherever gauges are used jointly, while QDM remains adequate for single-site work.

Keywords: multivariate bias correction; MBCn; inter-station dependence; multi-site precipitation; CMIP6

1. Introduction

Global climate models remain too coarse to be used directly at the scale of a catchment, and their daily rainfall carries systematic errors in both how often it rains and how hard, shortcomings that have proved stubborn across successive model generations [1]. Correcting these fields by mapping the simulated distribution onto the observed one is now standard practice [2,3], and among the available transformations quantile delta mapping (QDM) has become something close to a default. Its appeal is specific: it repairs the whole distribution at a site while transmitting, without alteration, the relative change the model projects in each quantile [4]. Where the end product is read at one gauge, little more is required.

What QDM does not do, in common with almost every method in routine use, is exercise any control over the relationships between gauges. Each site is corrected in isolation, so whatever the transformation does to

the joint structure of the network happens incidentally. For a point estimate this is of no consequence. It matters as soon as the corrected fields are fed to a distributed hydrological model, used to estimate flood frequency jointly across a basin, or used to delineate the area affected by a drought or a storm, because in those settings the behaviour of the field as a whole is the object of interest. Marginal distributions can be reproduced faultlessly while the joint structure is misrepresented, and any model downstream will inherit that error.

A separate line of work targets the joint distribution explicitly, including correlation-matching schemes, rank-based resampling [5], corrections formulated through optimal transport [6], and the N-dimensional distribution transform known as MBCn [7]. What these approaches share is an ability to reinstate dependence that univariate mapping leaves alone, and comparative work has shown that the choice between the two families can shift the result of a multivariate impact calculation appreciably [8]. MBCn extends quantile mapping into several dimensions simultaneously: the correlated variables are rotated through a random orthogonal frame, each rotated axis is quantile-mapped onto the equivalently rotated observations, the rotation is undone, and the cycle repeats until the joint distribution has been drawn toward the observed one, whereupon the station-level marginals from an initial QDM pass are reinstated.

That closing step determines how any comparison between the two methods must be interpreted. Reinstating the QDM marginal at every gauge forces the two corrections to return identical values on any distribution-based single-site score. A study reporting only such scores will find nothing to choose between them and may conclude that the multivariate machinery is superfluous. Testing whether it earns its cost therefore requires diagnostics aimed at the dimension it acts upon, namely the dependence among gauges, which a marginal-only assessment is structurally incapable of seeing.

Rainfall extremes and their spatial organisation are among the more consequential and less certain aspects of regional climate change [9], and CMIP6-based assessments for Thailand and the wider region have received sustained attention [10,11]. Studies that combine a genuinely withheld validation period with a broad set of dependence diagnostics inside a single station-network design nonetheless remain scarce, and it is this evaluation gap that motivates the present work. Working with twelve complete gauge records on the western Gulf of Thailand coast and a seven-model CMIP6 ensemble, the study poses one question: once single-site performance is held fixed by construction, how much does a dependence-aware correction improve the joint behaviour of the network, and does it pay for that improvement anywhere else? Three steps answer it. The marginal equivalence of the two methods is verified rather than assumed; the dependence structure is then examined through six complementary diagnostics with a paired test across models; and the numerical behaviour of the iterative step, together with its effect on the projected change signal, is checked to establish that the improvement is not bought at a hidden cost.

2. Materials and Methods

2.1 Study area and rain-gauge network

Twelve daily rain gauges in Phetchaburi and Prachuap Khiri Khan provinces, on the western shore of the Gulf of Thailand, provide the observational basis (Table 1, Figure 1). Each record runs unbroken from 1 January 1981 to 31 December 2014, thirty-four complete years with no missing day at any site. The network is deliberately varied rather than homogeneous: annual totals span roughly 944 to 1,376 mm, and the

proportion of wet days ranges from a little over one in five to nearly two in three. Contrasts of this size make the network a searching test, since a correction indifferent to space has ample room to smooth them away.

Coordinates and elevations come from the project metadata archive and are reported as recorded. Figure 1 locates the gauges inside the provincial boundary, taken from a shapefile in UTM Zone 47N and reprojected for display. The shading is an inverse-distance surface of mean annual rainfall included for orientation and clipped to the study area; no quantitative result depends on it.

Table 1. Characteristics of the twelve rain gauges, 1981–2014.

Station	Latitude (°N)	Longitude (°E)	Elevation (m)	Mean annual rainfall (mm)	Wet-day frequency (%)
500001	12.370	99.950	8	983.7	22.23
500002	11.180	99.550	0	1,376.2	33.73
500003	12.020	99.900	0	1,100.0	22.93
500004	11.470	99.650	0	1,106.0	26.34
500005	11.600	99.680	6	1,147.2	47.96
500006	11.620	99.670	24	1,203.1	49.77
500007	11.900	99.780	24	1,133.6	48.66
500008	12.530	99.550	128	1,206.3	62.63
500009	12.220	99.870	13	1,176.5	60.15
500201	11.830	99.830	64	1,108.2	24.70
500202	12.590	99.960	0	944.1	21.53
500301	12.580	99.730	106	1,064.8	25.33

Coordinates (WGS84) and elevations are from the project metadata archive; rainfall statistics are computed from the gap-free observed daily record. A wet day is defined throughout as at least 1 mm of rainfall.

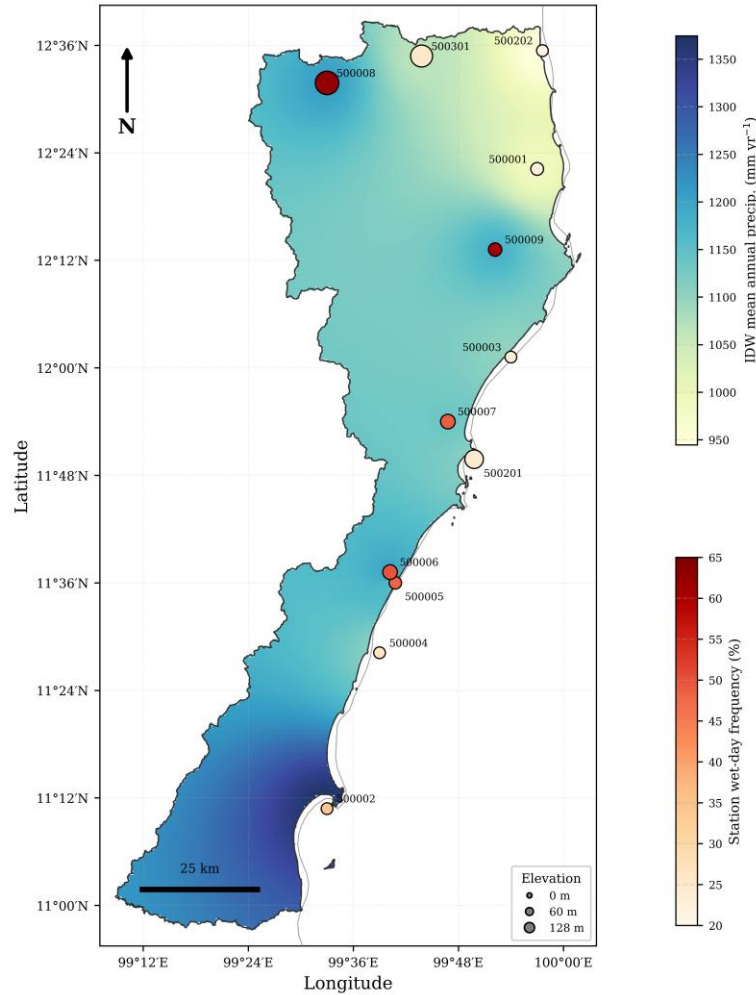


Figure 1. The twelve-gauge network on the Phetchaburi–Prachuap Khiri Khan coast. Marker size scales with elevation and fill with wet-day frequency. The shaded background is an inverse-distance surface of mean annual rainfall, shown for orientation only and clipped to the provincial boundary (reprojected from UTM Zone 47N).

2.2 Climate models and experimental design

Seven CMIP6 models supply the simulated rainfall (Table 2), each with a continuous daily record at the gauge locations spanning the historical experiment and both future pathways. Ensemble member and grid labels were taken from the archive filenames rather than from secondary documentation, which is how the r11i1p1f1 realisation of CESM2 was distinguished from the r1i1p1f1 realisations used for the remaining six models.

The record was split once and that split governed every method. Training used the two decades 1981–2000. The fourteen years from 2001 to 2014 were held back completely and touched only at the scoring stage, so that no difference in skill can be attributed to differences in the information available to a method. Only gauges present in the observed record define the network; any additional sites appearing in the model files were discarded, confining every comparison to the same twelve locations.

Table 2. The seven CMIP6 models and their configuration.

Model	Institution	Ensemble member	Grid label
ACCESS-ESM1-5	CSIRO, Australia	r1i1p1f1	gn
CanESM5	CCCma, Canada	r1i1p1f1	gn
CESM2	NCAR, United States	r1i1p1f1	gn
EC-Earth3	EC-Earth Consortium, Europe	r1i1p1f1	gr
FGOALS-g3	CAS, China	r1i1p1f1	gn
MIROC6	MIROC, Japan	r1i1p1f1	gn
MRI-ESM2-0	MRI, Japan	r1i1p1f1	gn

All seven models provide daily precipitation for the historical experiment (1981–2014) and for both SSP2-4.5 and SSP5-8.5. Labels were verified against the archive filenames. gn = native model grid; gr = regridded.

2.3 Bias-correction methods

QDM operates on one gauge and one calendar month at a time. Observed and simulated quantile functions are estimated over the training window, applied at whatever quantile a given value occupies in the target window, and combined so that the model's own relative change is carried through intact, neither amplified nor suppressed [4]. MBCn takes the QDM-corrected field as its starting point and adds an iterative stage [7]. Each pass draws a random orthogonal rotation spanning the twelve station dimensions, quantile-maps every rotated component onto the correspondingly rotated observations, then reverses the rotation. A fixed random seed determines the sequence of rotations, which is shared between the training and target windows. After twenty-five passes, rank substitution restores the QDM marginal at each station, leaving a field with QDM's single-site distributions and an inter-station rank structure drawn toward the observations (Figure 2).

Every estimated quantity, including the QDM quantile maps, the rotation sequence, and the quantile maps applied in rotated space, was fitted on the 1981–2000 data alone. Observations from 2001–2014 played no part in fitting and were used only to score the transformed series. The dependence target is the training-period observed structure, so no information passes from the evaluation window into the correction. A wet day is defined as at least 1 mm of rainfall throughout, applied identically to observed and corrected series. Percentile thresholds used in the extreme diagnostics are likewise computed within each series from its own wet days, so that those diagnostics respond to dependence structure rather than to any residual difference in intensity between series.

Conceptual workflow — MBCn bias correction
(orange = marginal correction step; blue = iterative dependence-alignment core;
green = final MBCn output preserving QDM marginals)

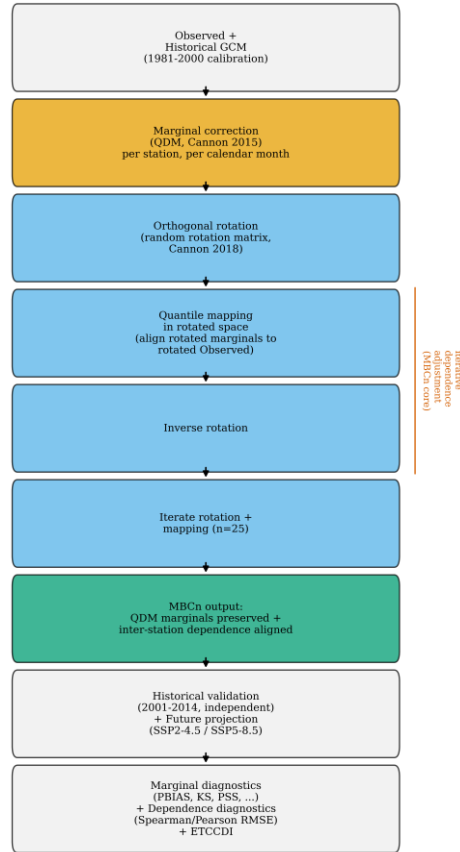


Figure 2. The MBCn workflow. An initial QDM step (orange) fixes the marginal at each station; the iterative rotate–map–unrotate core (blue) aligns the inter-station dependence; the QDM marginal is reinstated at the end (green). Fitting uses training data only.

2.4 Single-site diagnostics

Free-running climate models are not synchronised with observed weather on any particular date, so single-site skill is assessed distributionally rather than day against day. Six scores were computed for each of the eighty-four gauge–model combinations over the withheld window: percentage bias in total rainfall, the error in wet-day frequency, the ratio of corrected to observed wet-day standard deviation, the Perkins score measuring overlap between wet-day densities [12], the Kolmogorov–Smirnov distance, and the root-mean-square error between matched wet-day quantiles. Day-matched criteria such as Kling–Gupta or Nash–Sutcliffe efficiency were avoided as headline measures, since they penalise a free-running simulation for timing it was never designed to reproduce.

2.5 Dependence diagnostics and statistical testing

Six diagnostics address the joint behaviour of the network from complementary directions. The broadest is the energy distance between the twelve-dimensional daily rainfall vectors of a method and of the observations, a two-sample statistic sensitive to the whole joint distribution rather than to its second

moments alone [13]. To keep the pairwise distance computation tractable it is evaluated on a random subsample of at most 1,500 days, drawn without replacement with a fixed seed from the roughly 5,100-day validation window; the subsampling introduces a small positive offset that is identical across methods and cancels in the paired comparison. Three further measures take the root-mean-square error over the sixty-six off-diagonal station pairs between the observed correlation matrix and that of each method, using Spearman rank correlation, Pearson correlation, and the correlation of the binary wet–dry indicator. Two diagnostics target extremes: the error in the pairwise probability that both gauges exceed their own ninety-fifth wet-day percentile on the same date, and the discrepancy in the distribution of how many gauges are wet simultaneously, summarised by a Wasserstein-1 distance. Every diagnostic is computed separately for each model.

The central comparison rests on the seven paired model-level differences between MBCn and QDM, assessed with a one-sided Wilcoxon signed-rank test and reported alongside the matched-pairs rank-biserial effect size and the number of models favouring MBCn. Seven models neither justify a normality assumption nor constitute a random sample of any defined population, so the test is presented as a formal summary of a paired improvement rather than as inference about climate models generally. A Holm–Bonferroni adjustment is applied within the pre-declared family of three correlation endpoints, namely Spearman, Pearson, and wet-day co-occurrence, so that the standing of the primary Spearman endpoint does not rest on the remaining diagnostics, which are treated as exploratory.

2.6 Robustness checks

Two properties were verified before any conclusion was drawn. Sensitivity to the iterative procedure was examined by varying the number of passes from zero, which recovers pure QDM, up to one hundred, and by repeating the correction with twenty different random seeds at the default setting. Fidelity to the projected change signal was assessed by comparing the percentage change in five wet-day quantiles between the historical and future windows under the raw model against the same change after each correction, evaluated for every gauge, model, pathway, and future slice. This check concerns the quantile change alone; it makes no claim about the projections themselves, which lie outside the scope of this evaluation.

3. Results and Discussion

3.1 Single-site performance is identical between the two corrections

Both corrections repair the familiar deficiencies of the raw output, which rains on far too many days, with wet-day frequency about eleven percentage points too high, and produces wet-day intensities that are too uniform, giving a standard-deviation ratio of 0.69 (Table 3, Figure 3). After correction the Kolmogorov–Smirnov distance falls from 0.221 to 0.101 and the Perkins overlap rises from 0.82 to 0.91. A residual dry bias of about seventeen percent in total rainfall remains, accompanied by a wet-day frequency that is now some six percentage points too low; the two are linked, since a correction that reduces the number of raining days necessarily removes volume, and the residual should be borne in mind when the corrected series are used for water-balance purposes.

QDM and MBCn returned the same value to three decimal places on every score and every one of the eighty-four gauge–model combinations. Far from being anomalous, this is the algorithm behaving exactly as specified: the closing step of MBCn reinstates the QDM value at each gauge, so both methods operate on an identical set of daily values at every site and cannot disagree on any distribution-based statistic, nor on bias, spread, or wet-day frequency. The proper inference is not that the multivariate stage accomplishes nothing but that single-site scores are the wrong place to look for its effect. Verifying the equivalence empirically, rather than inferring it from the algorithm, also establishes that the implementation behaves as intended, and it isolates the joint structure as the only remaining axis on which the methods can differ.

Table 3. Single-site skill over the withheld validation period (mean $\pm$ SD, $n = 84$ gauge–model pairs).

Method	PBIAS (%)	Wet-day frequency error (pp)	SD ratio (–)	PSS (–)	KS distance (–)	Quantile RMSE (mm d ^{–1})
RAW	1.4 $\pm$ 34.7	10.6 $\pm$ 17.2	0.69 $\pm$ 0.49	0.82 $\pm$ 0.10	0.221 $\pm$ 0.104	7.83 $\pm$ 4.16
QDM	–17.2 $\pm$ 20.8	–6.4 $\pm$ 12.3	0.98 $\pm$ 0.36	0.91 $\pm$ 0.05	0.101 $\pm$ 0.060	4.08 $\pm$ 3.05
MBCn	–17.2 $\pm$ 20.8	–6.4 $\pm$ 12.3	0.98 $\pm$ 0.36	0.91 $\pm$ 0.05	0.101 $\pm$ 0.060	4.08 $\pm$ 3.05

pp = percentage points; PSS = Perkins skill score; KS = Kolmogorov–Smirnov; SD ratio is the corrected-to-observed ratio of the wet-day standard deviation. QDM and MBCn coincide on every score because MBCn reinstates the QDM value at each gauge.

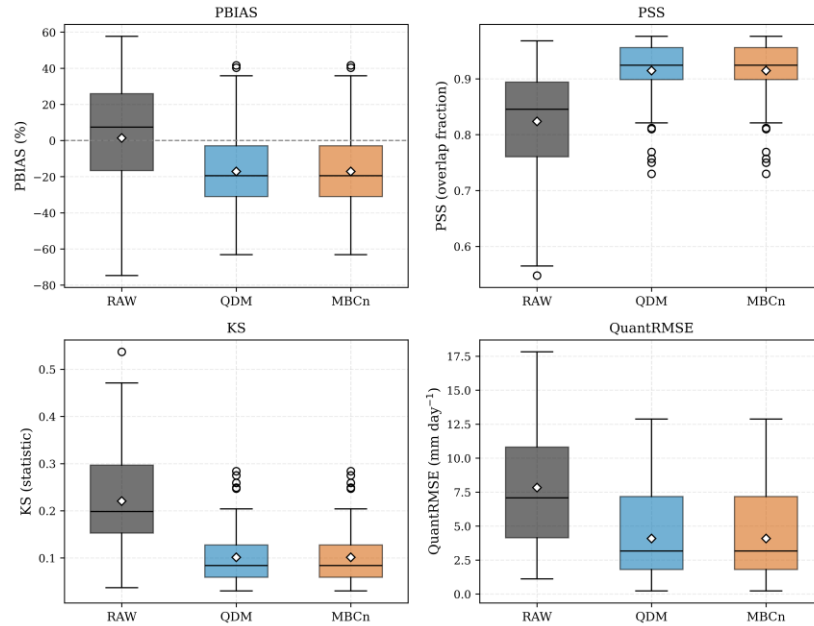


Figure 3. Single-site skill for RAW, QDM, and MBCn over the withheld validation period, one point per gauge–model pair ($n = 84$). Diamonds mark the mean. The QDM and MBCn distributions coincide exactly, by construction.

3.2 MBCn improves inter-station dependence consistently across models

Once attention moves to the joint structure the two methods separate on every diagnostic examined (Table 4). Taking the joint distribution as a whole through the energy distance, MBCn reaches 0.571 against 0.852 for QDM, closing a third of the remaining gap to the observations. The correlation structure agrees: the Spearman-matrix error, already reduced from 0.484 in the raw output to 0.286 by QDM, falls again to 0.187

under MBCn. Univariate correction therefore removes a substantial part of the spurious dependence the raw model imposes, and the multivariate stage removes more than half of what survives. Pearson correlation, wet-day co-occurrence, joint exceedance at the ninety-fifth percentile, and the network-wide wet-extent distribution all point the same way.

The weight of this result comes from its consistency rather than the size of any single number. On all six diagnostics, every one of the seven models lies closer to the observations under MBCn than under QDM (Table 5). Such unanimity produces the largest matched-pairs rank-biserial effect size attainable and drives the one-sided Wilcoxon p-value to 0.0078, the minimum available with seven pairs; Holm–Bonferroni adjustment within the three correlation endpoints leaves each at 0.0234, so the pre-declared Spearman endpoint clears the conventional threshold without support from the secondary diagnostics. Reporting the seven-of-seven count alongside the p-value is deliberate, since a p-value derived from seven observations carries limited weight on its own. The interpretive limit is worth restating: seven models are not a random sample, and the test summarises a unanimous, evenly sized paired improvement rather than supporting any claim about climate models at large. It is also worth noting that the six diagnostics are computed from the same daily series and are strongly interrelated, so they constitute six views of one improvement rather than six independent confirmations.

Nothing in the algorithm guarantees this outcome, which is what makes it informative. MBCn is constructed to move a joint distribution toward a target, but whether the mechanism succeeds uniformly across models differing in resolution and convective parameterisation is an empirical matter, and here it does. Figures 4 and 5 display the result from two angles. Figure 4 shows the inter-station correlation matrices for MIROC6, chosen in advance as the model whose MBCn error sits closest to the ensemble median so that the panel is representative rather than flattering. The raw field assigns a near-uniform high correlation to most gauge pairs; QDM lightens this only partially; MBCn recovers a texture much closer to the observed, including the genuinely weak links the raw model erases. Figure 5 retains all seven models, each thin line connecting a model's QDM and MBCn score for one diagnostic, and every line in every panel slopes downward.

Figure 6 avoids the shelter of an average by plotting the Spearman correlation of each gauge pair against its separation. Observed correlation declines gently with distance, from roughly 0.47 at the shortest separations to 0.39 at the longest, the signature of a convective coastal regime in which storm cells lose coherence over tens of kilometres. The raw output replaces this decay with an almost flat coherence near 0.96, which makes the over-coherence concrete; QDM reduces the near-pair values without bringing them far enough down; MBCn follows the observed decay most closely. Part of the raw behaviour is structural rather than a bias in the ordinary sense, since several gauges are extracted from the same or adjacent coarse grid cells, and same-cell extraction imposes an artificial coherence that a univariate method cannot detect and only a dependence-aware one can undo. The binned relationship is descriptive, because the sixty-six pairs are not independent and the bootstrap band is therefore exploratory, but the ranking of the three methods is unambiguous.

Table 4. Dependence diagnostics against observations (seven-model mean, withheld validation period). All scores are negatively oriented, so lower is better.

Method	Energy distance	Spearman RMSE	Pearson RMSE	Wet-occurrence RMSE	Joint q95 RMSE	Wet-extent W_1
RAW	1.283	0.484	0.488	0.476	0.0150	0.174
QDM	0.852	0.286	0.374	0.273	0.0054	0.107
MBCn	0.571	0.187	0.134	0.139	0.0029	0.079

Values are means of per-model scores. The energy distance is the broadest whole-distribution measure, computed on a fixed-seed random subsample of at most 1,500 days; the correlation-matrix RMSEs, the joint-exceedance RMSE and the wet-extent Wasserstein-1 distance (W_1) are structure diagnostics. Joint exceedance uses each series' own wet-day ninety-fifth percentile and wet extent the 1 mm threshold. MBCn is closest to observations on every diagnostic. All quantities are dimensionless.

Table 5. Paired comparison of MBCn against QDM across the seven models (one-sided Wilcoxon signed-rank test; Holm-Bonferroni adjustment within the correlation-endpoint family).

Diagnostic	Models favouring MBCn	W	p	Holm-adjusted p	Effect size
Energy distance (secondary)	7 / 7	28.0	0.0078	–	1.00
Spearman RMSE (primary)	7 / 7	28.0	0.0078	0.0234	1.00
Pearson RMSE	7 / 7	28.0	0.0078	0.0234	1.00
Wet-day co-occurrence RMSE	7 / 7	28.0	0.0078	0.0234	1.00
Joint q95 exceedance RMSE (secondary)	7 / 7	28.0	0.0078	–	1.00
Wet-extent W_1 (secondary)	7 / 7	28.0	0.0078	–	1.00

Every model improves under MBCn on every diagnostic, yielding the maximum matched-pairs rank-biserial effect size and the smallest p attainable at $n = 7$. The Holm adjustment covers the three pre-declared correlation endpoints; the remaining diagnostics are exploratory and reported unadjusted.

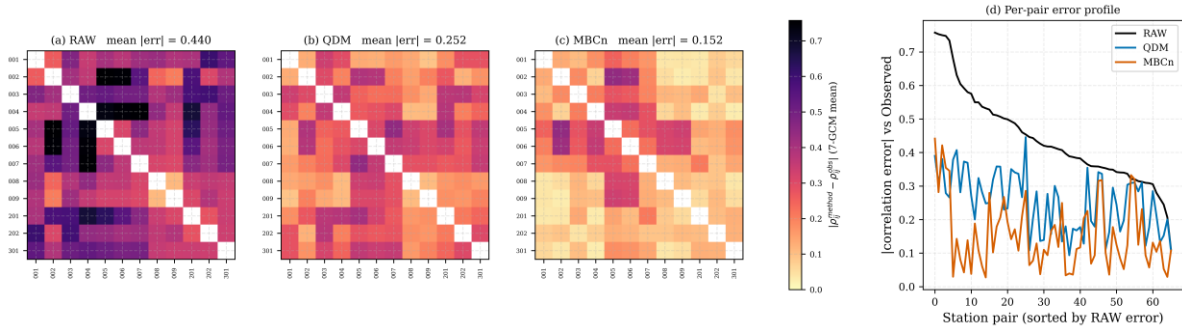


Figure 4. Inter-station Spearman correlation for the median-performing model (MIROC6, selected before inspection). RAW imposes a near-uniform high correlation, QDM corrects it only partially, and MBCn recovers a texture close to the observed. Panel (d) shows the sorted per-pair error.

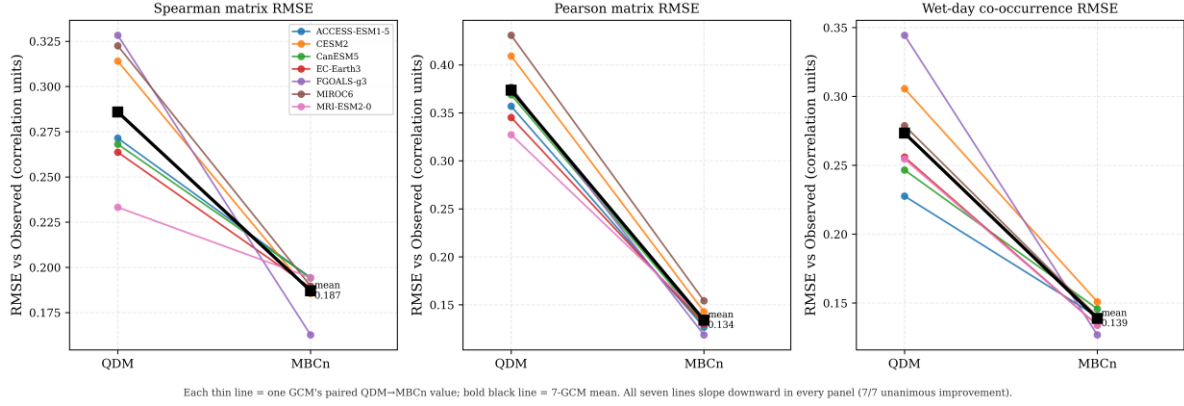


Figure 5. Paired QDM→MBCn dependence error for each of the seven models (thin lines) and the ensemble mean (bold), for three diagnostics. Every line slopes downward, the graphical form of the 7/7 result in Table 5.

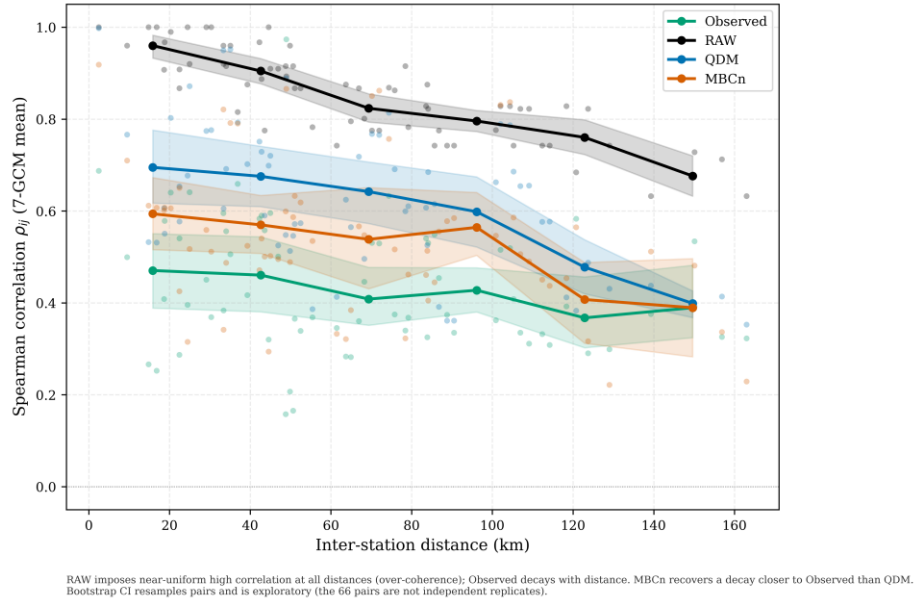


Figure 6. Inter-station correlation against separation distance. Observations decay gently, RAW remains near-uniformly high, and MBCn follows the observed decay more closely than QDM. Bootstrap bands are exploratory because gauge pairs are not independent.

3.3 Spatial organisation of concurrent extremes

Correlation matrices summarise dependence but reveal little about the events that matter hydrologically, so two diagnostics were framed in terms of concurrence. Figure 7 compares, for each gauge pair, the probability that both sites record a top-five-percent wet day on the same date. Raw output scatters these probabilities well away from the one-to-one line; QDM draws them in but leaves a visible positive bias; MBCn brings them closest to observed and alone shows no systematic tilt. The improvement is unanimous across models (Table 5), which is what permits the language of spatial coherence rather than correlation structure alone.

Figure 8 restates the question in a form a hydrologist can act on: how much of the network is wet on a given day. The ordering follows the pattern already established, with MBCn closest to the observed distribution

and every model improving relative to QDM (Tables 4 and 5). Because the diagnostic counts stations rather than millimetres, it translates the abstract notion of dependence into the distinction between a local shower, a subregional event, and one that covers the whole network, which is the distinction a distributed rainfall-runoff model responds to. The equivalent distribution for days in the upper tail is provided as Supplementary Figure S1; it is pooled across models and uses observed intensity thresholds, so the paired model-level test does not cover it and it is not interpreted here.

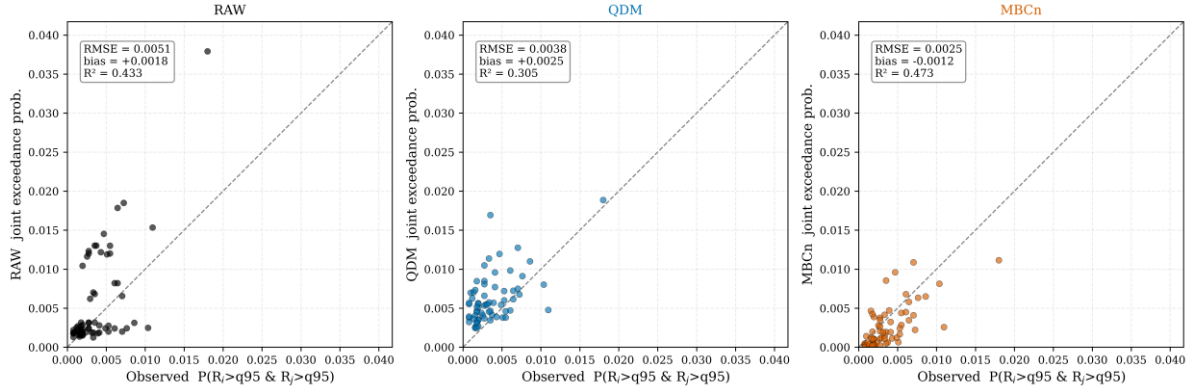


Figure 7. Pairwise probability that both gauges exceed their ninety-fifth wet-day percentile on the same day, observed against corrected (seven-model mean). MBCn lies closest to the 1:1 line and is the only method without a systematic tilt.

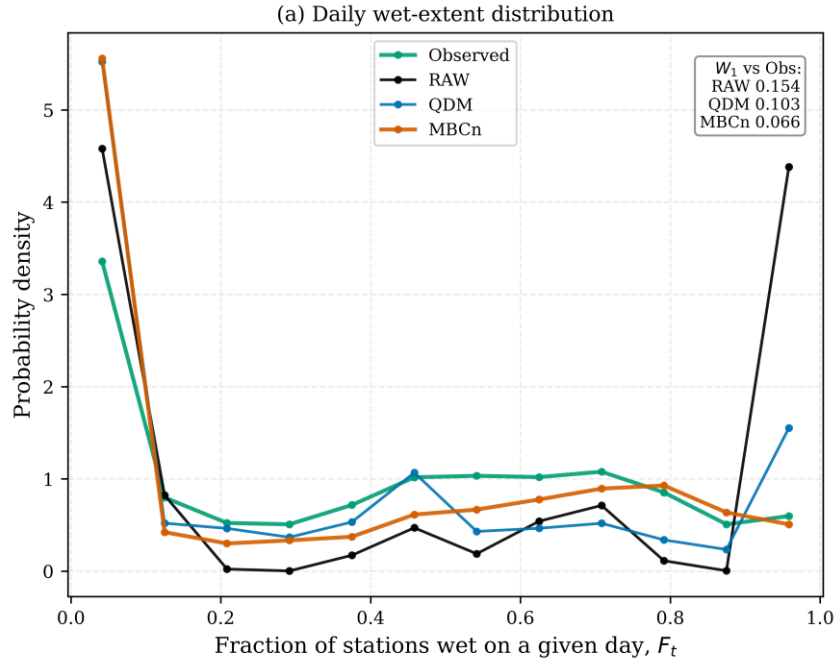


Figure 8. Distribution of the number of gauges simultaneously wet, observed against each method, pooled across the seven models. This is the wet-extent diagnostic reported per model in Tables 4 and 5.

3.4 Numerical behaviour and fidelity to the projected change signal

Two checks establish that the multivariate stage is well behaved and carries no concealed cost. Figure 9 follows three dependence scores as the number of passes rises from zero, the pure QDM starting point, to

one hundred. Most of the gain is realised within roughly ten to fifteen passes, after which the diagnostics oscillate within a narrow band, placing the default of twenty-five comfortably on the plateau rather than at an arbitrary cut-off. Repeating the correction with twenty random seeds at that default leaves the correlation-based scores essentially unchanged, with a coefficient of variation near four percent. The wider scatter in the energy score reflects the random subsampling that makes the statistic affordable rather than instability in the algorithm, which is why robustness is judged from the full-sample correlation diagnostics.

Figure 10 addresses the objection most likely to be raised against any method that reorders days, namely that rearrangement might blur the change the model projects. Plotting the projected change in five wet-day quantiles under the raw model against the same change after correction shows QDM and MBCn tracking the raw signal almost interchangeably: the mean departure under MBCn is 3.7 percentage points for the moderate pathway and 7.3 for the high one, against 4.1 and 7.5 for QDM. Whatever change a model projects therefore survives the multivariate correction as intact as it survives QDM, and the dependence gain documented above is obtained without sacrificing the property for which QDM is valued [4]. The check covers quantile change only and offers no assurance for statistics that depend on the sequencing of wet and dry spells, such as run-length indices, which by construction MBCn may alter.

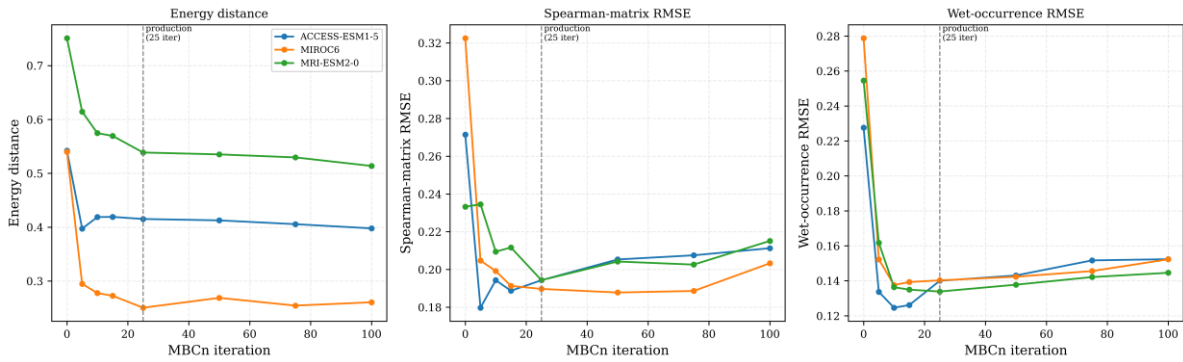


Figure 9. Dependence scores against MBCn iteration count (iteration 0 is the QDM starting point). Most of the improvement occurs within roughly ten to fifteen passes, placing the default of twenty-five on the plateau.

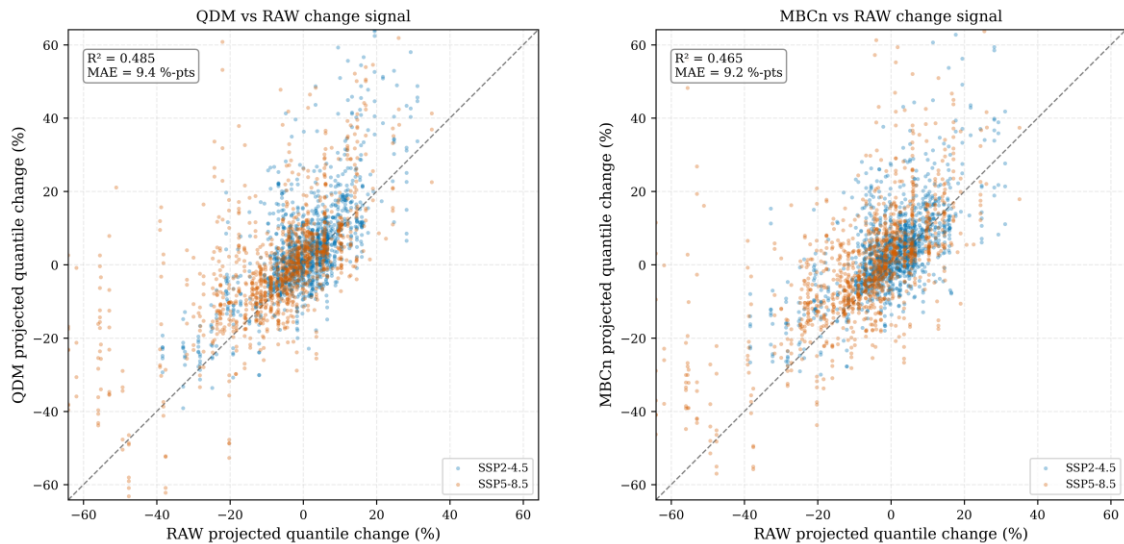


Figure 10. Projected wet-day quantile change under the raw model against the same change after each correction, for every gauge, model, pathway, and future slice. MBCn tracks the raw change as faithfully as QDM.

3.5 Implications and limitations

Taken together the results support a decision rule rather than a ranking. Where corrected rainfall is used one gauge at a time, as in a design storm, an intensity–duration–frequency curve, or a single-site trend, QDM is sufficient, cheaper, and statistically indistinguishable from MBCn on everything that matters in that context. Where gauges are used jointly, as in distributed hydrological modelling, joint flood-frequency estimation, or the mapping of drought and flood extent, the evidence favours MBCn, and the extra computation buys a measurable and unanimous gain in the realism of the spatial field. The change-signal check removes the main reason for hesitation. These findings sit alongside the multivariate-correction literature [7,8] and extend it by demonstrating consistency across a structurally diverse ensemble under a strictly withheld validation design.

A broader methodological point follows. If two corrections can be indistinguishable on every single-site score yet differ substantially in joint behaviour, then station-wise evaluation cannot separate methods on a property many downstream applications depend on. Reporting at least one dependence diagnostic alongside the customary marginal scores would cost little and would make such differences visible.

Four limitations bound these conclusions. One realisation per model means the ensemble spread represents structural differences between models but not internal variability, which would widen the envelopes. The network is a single coastal basin, complete and spatially distributed but not necessarily representative; whether the advantage holds in a flatter interior or a more strongly monsoonal setting is a fair question this study cannot settle. The correction is trained on a window treated as stationary, so it restores the observed twentieth-century dependence structure and cannot represent a change in inter-station coherence should one occur. Finally, the residual dry bias common to both corrections limits the use of these series for water-balance work and would confound any diagnostic that applies observed intensity thresholds to corrected output. None of these bears on the central comparison, which concerns two corrections applied to identical data, but each constrains how far the result should be generalised.

4. Conclusion

Compared side by side on twelve complete coastal gauge records and a seven-model CMIP6 ensemble, QDM and MBCn prove indistinguishable to any single gauge, a built-in consequence of how MBCn reinstates its marginals, and clearly distinct wherever the gauges are considered together. Across a multivariate energy distance, three correlation structures, the joint exceedance of heavy rain at gauge pairs, and the day-to-day extent of wet conditions, MBCn moves consistently closer to observations, unanimously across all seven models, while converging within roughly fifteen passes, proving insensitive to the random seed, and preserving projected quantile changes as faithfully as QDM. The finding argues not for universal adoption of multivariate correction but for matching the method to the question: QDM remains the efficient default where only marginal fidelity is required, whereas MBCn is the better-supported choice wherever spatial coherence forms part of what is being asked, as it commonly does in hydrological and flood applications. More generally, the demonstration that two corrections can be identical on every single-site score yet differ substantially in joint behaviour suggests that dependence diagnostics belong in the routine

evaluation of bias-correction methods. Extending the same paired design to a larger and topographically more varied network, and to multi-realisation ensembles, would establish how far the magnitude of the advantage reported here generalises.

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Author Contributions

Conceptualization: X.X.; Methodology: X.X.; Software: X.X.; Validation: X.X. and Y.Y.; Formal analysis: X.X.; Investigation: X.X.; Resources: Y.Y.; Data curation: X.X.; Writing—original draft preparation: X.X.; Writing—review and editing: X.X. and Y.Y.; Visualization: X.X.; Supervision: Y.Y.; Project administration: X.X. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest

The authors declare no conflict of interest.

Data and Code Availability

The bias-correction, evaluation, and figure code is deterministic given a fixed random seed and is organised so that a new study area can be configured through a single settings file. Each figure is accompanied by the numeric table underlying it. The iterative implementation used for the convergence and seed analyses was verified to reproduce the reference engine's output exactly before any sensitivity value was accepted. Daily rain-gauge records are available from the Royal Irrigation Department subject to its data-sharing conditions; CMIP6 output is publicly available through the Earth System Grid Federation. Supplementary Figure S1 is provided with the submission.

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